What makes SWE be different other engineering branches is product intangibility (they are really easy to change)

SWE PROCESSES

We have: Specification, Development, Validation & Verification, Evolution

- Specification includes requirements
- Development includes coding design
- Valid am I doing right product?
- Verification am I doing product right?
- First validation then verification
- Evolution: Maintenance of product

Waterfall: seperate phases of specification and development.
Evolutionary Development: specification, development and validation are interleaved.
Component based swe: The system is assembled from existing components.
Exploratory Development: Initial prototype ~~> evolve ~~> final sytem needs to start well understood requirements.
Throw-away Prototyping: It should start with poorly understood requirements. objective is to understan requirements.
Evolutionary Development includes explotary development and throw-away

Commercial-off-the-shelf (existing components) it is based on reuse of existing components.
component analysis
requirements modification
system design with reuse
development and integration

Process Iteration has two approches:
- Incremental delivery
- Spiral development

*Incremental delivery is not single delivery, user requirements are prioritised and the higher the priority of a requirement the sooner it is delivered. Extreme programming is derivative of incremental development and it relies on pair programming.

* Spiral Development has no fixed phases such as specification - design.
Loops are chosen depending on what is required

Feasibility study is first requirements engineering process where the question of "is it worth to work on project?"

Software design: Design of a software structure - model
Implementation: Translation of design to an executable program

unit testing: individual components are tested
sytem testing: testing the system as a whole
acceptance testing: testing with customer

RUP = Inception (like specification) + elaboration (understanding problem domain) + construction (design programming) + transition (deploying)
CASE (computer-aided software engineering): diagram tools + version management system + UI generators + compilers etc.

REQ. ENGINEERING

User Requirements = written for customers + diagram of services of the system
System Requirements = detailed desc of system's functions
There is another classification as functional, nonfunctional, domain requirements.

Functional requirements are classified into:
- Functional User Requirements
- Functional System Requirements

User Requirements are ambiguous.

Completeness: includes descriptions of all facilities required.
Consistency: no conflicts or contradictions

Nonfunctional requirements are more critical than functional requirements. Classification
- Product requirements
How Product should behave?
- Organisational requirements
Process standards-policies etc.
- External Requirements (legislative req)
Factors outside of the system, government...

Nonfunctional should have measure...
My system should respond fast → Not verifiable nonfunc requirement
The system should respond in 2 sec → VERIFIABLE non functional requirement

Conflicts between Nonfunctional requirements are possible in complex systems. Requirements prioritizations may be needed.

Domain requirements describe system characteristics reflecting the domain. Understandability and implicitness can be problem.

Requirements amalgamation multiple requirements mixed and pointing out a single requirement. Ambiguity)

Alternatives to NL specification includes: (Natural Language) Structured natural language, design description language Graphical Notations (UML), Mathematical Specification

At what phase, maintenance engineers active? Evolutionary phase.

Feasibility study checking usefullness of system on customers viewpoint (worth to work on?)
Elicitation and analysis involves staff working with customers to find out about the application domain, services, constraints...

The requirements spiral has phases like Requirements Discovery, Requirements classification & organization Prioritisation & negotiation, Requirements Documentation

Enduring Requirements: stable requirements core of the domain.
Volatile Requirements: may change when the system is in use.

Traceability is concerned with the relationships between requirements, their sources and the system design.
R: weak dependency D: strong dependency

	1	2	3	4	5
1			R		
2		R		D	
3					

CamScanner ile tarandı